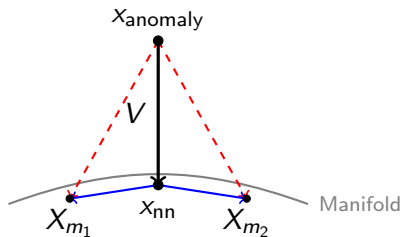


1. The Geometric Anchor

$$G = \sum_{m=1}^K P_{x \rightarrow m} \Delta_m \Delta_m^T$$

$$V = V_{\text{tan}} + V_{\text{ortho}}$$



$$\underbrace{\Delta_m = X_m - x_{\text{anomaly}}}_{\text{Ambient Cone}} \Rightarrow V_{\text{ortho}} \rightarrow 0$$

$$\underbrace{\Delta_m = X_m - x_{\text{nn}}}_{\text{Tangent Plane}}$$

2. The 1D Covariance Collapse (Original Flaw)

$$V_{\text{tan_raw}} = \sum_{m=1}^K P_{x_{\text{nn}} \rightarrow m} (V \cdot \Delta_m) \Delta_m$$

$$V_{\text{tan_1D}} = \left(V \cdot \frac{V_{\text{tan_raw}}}{\|V_{\text{tan_raw}}\|} \right) \frac{V_{\text{tan_raw}}}{\|V_{\text{tan_raw}}\|}$$

$$V_{\text{tan_raw}} = GV = \sum_{i=1}^K \underbrace{\sigma_i^2}_{\text{Variance Scaling}} (V \cdot w_i) w_i$$

$$\underbrace{V_{\text{tan_1D}}}_{\text{Rank 1}} \neq \underbrace{\sum_{i=1}^d (V \cdot w_i) w_i}_{\text{Rank } d}$$

3. SVD Bypass: Extracting the Orthonormal Basis

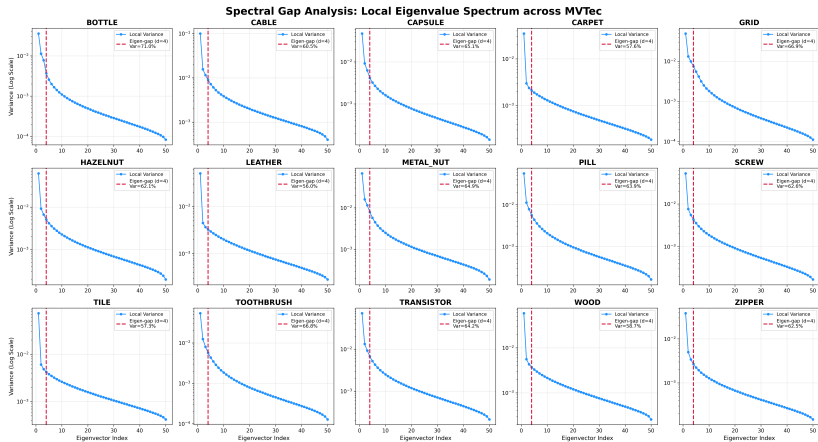
$$\tilde{X} = \begin{bmatrix} \sqrt{P_{x_{nn} \rightarrow 1}} \Delta_1^T \\ \vdots \\ \sqrt{P_{x_{nn} \rightarrow K}} \Delta_K^T \end{bmatrix} \in \mathbb{R}^{K \times D}$$

$$\tilde{X} = U \Sigma W^T$$

$$G = \tilde{X}^T \tilde{X} = (W \Sigma U^T)(U \Sigma W^T)$$

$$G = W \Sigma^2 W^T \Rightarrow W = [w_1, w_2, \dots, w_K]$$

4. No Eigen-Gap



5. Dynamic Intrinsic Dimension (Variance Masking)

$$\lambda_i = \sigma_i^2$$

$$R_k = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^K \lambda_j}$$

$$d = \min \{k \mid R_k \geq 0.90\}$$

$$V_{\text{tan}} = \sum_{i=1}^d (V \cdot w_i) w_i$$

$$V_{\text{ortho}} = V - V_{\text{tan}}$$

6. Patchcore CDC Results

Category	IM224 WR-50 (Baseline)	IM320 Ensemble	Improvement
bottle	1.0000	1.0000	0.0000
cable	0.9968	0.9979	+0.0011
capsule	0.9792	0.9868	+0.0076
carpet	0.9859	0.9963	+0.0104
grid	0.9791	0.9949	+0.0158
hazelnut	1.0000	1.0000	0.0000
leather	1.0000	1.0000	0.0000
metal_nut	0.9990	1.0000	+0.0010
pill	0.9667	0.9833	+0.0166
screw	0.9877	0.9924	+0.0047
tile	0.9949	0.9877	-0.0072
toothbrush	1.0000	1.0000	0.0000
transistor	0.9987	0.9987	0.0000
wood	0.9912	0.9973	+0.0061
zipper	0.9950	0.9968	+0.0018
Mean	0.9916	0.9955	+0.0039

7. Drift CDC Results

Category	PatchCore Baseline	CDC Optimized	Improvement
bottle	1.0000	1.0000	0.0000
cable	0.9968	0.9893	-0.0075
capsule	0.9792	0.9924	+0.0132
carpet	0.9859	0.9872	+0.0013
grid	0.9791	1.0000	+0.0209
hazelnut	1.0000	1.0000	0.0000
leather	1.0000	1.0000	0.0000
metal_nut	0.9990	1.0000	+0.0010
pill	0.9667	0.9861	+0.0194
screw	0.9877	1.0000	+0.0123
tile	0.9949	1.0000	+0.0051
toothbrush	1.0000	1.0000	0.0000
transistor	0.9987	0.9867	-0.0120
wood	0.9912	0.9912	0.0000
zipper	0.9950	0.9947	-0.0003
Mean	0.9916	0.9952	+0.0036